



DZPHARM · DESIGN SYSTEM

Color System

A detailed deep-dive into the ~400-token color architecture: 12 semantic token categories, 7-level state system, 20-palette Nuancier with cultural identities (Craie & Cobalt, Dune & Cèdre de l'Atlas, Perle & Azur d'Alger), dual-theme switching, 9-level shadow system with themed glows, WCAG AAA contrast on 11/12 pairs.

~400

CSS TOKENS

20

PALETTES

11/12

CONTRAST AAA

A

GRADE

DEEP-DIVE OF

Feature #53 – Color System

From the 68-item review menu

~400 CSS Custom Properties, 12 Categories

The color system is **the strongest design-system asset on the platform**. A probe of the rendered DOM revealed approximately **400 CSS custom properties** organized across 12 categories: background, foreground, card, popover, primary, secondary, muted, accent, destructive, border, input, and ring. Each category has both a dark-theme and light-theme value, swapped atomically when the theme toggles. This is senior-level design-system engineering, not the "developer-built" implementation that the earlier #5 Visual Design report claimed.

Nomenclature officielle — Juin 2026

L'intelligence pharmaceutique algérienne

Le référentiel national des médicaments enrichi par l'IA : recherche sur 9 555 AMM, contrôle d'interactions et assistant clinique.

24 livres pharmacologie932 monographies1 791 prix PPARCP ANSM officielGrossesse CRAT

PÉRIMÈTRE : TousDCI / MoléculeMarqueLaboratoireN° AMM

Rechercher en arabe ou darija (ex: بومادا صفراء, دوا السكر, باراسيتامول).ScannerEntrée

Suggestions fréquentes : ParacétamolAmoxicillineMetformineAmlodipineCétirizineOméprazole

Accès rapide

Sélection adaptée au mode professionnel

Répertoire
Rechercher un médicament par marque, DCI, laboratoire ou n° AMM.

Contrôle d'interactions
Vérifier les associations médicamenteuses avant dispensation.

RCP & monographies
911 monographies DCI issues des livres techniques de pharmacologie.

Calculs cliniques
Posologies pédiatriques, fonction rénale, grossesse, Chifa, Ramadan.

Copilote IA
Assistant clinique en français, arabe et darija.

MÉDICAMENTS ACTIFS
5 381
AMM en cours de validité

PRODUCTION LOCALE
3 840 71%
Part des actifs fabriqués en Algérie

IMPORTATIONS
1 541
Actifs importés

RETIRÉS DU MARCHÉ
2 679
Historique des retraits

Les plus consultés

Classement DzPharm — consultations cumulées

1 DOLIPRANE
PARACETAMOL
7 consultations

2 ACARLYSE
ACARBOSE
5 consultations

3 CLAVODAR FORTE 400/57
AMOXICILLINE (SOUS FORME TRIHYDRATEE)/ACI...
4 consultations

4 ABAVIR OS
ABACAVIR SULFATE EXPERIME EN ABACAVIR
3 consultations

5 BIOPAMOX
AMOXICILLINE SODIQUE
2 consultations

6 CO-DOLYC
PARACETAMOL/CODEINE PHOSPHATE HEMIHYD...
2 consultations

7 IMETH
METHOTREXATE
2 consultations

8 LEMBLASTINE
VINBLASTINE (SOUS FORME DE SULFATE)
2 consultations

DCI les plus recherchées

Classement par principe actif — consultations cumulées

1 PARACETAMOL
DOLIPRANE
9

2 ACARBOSE
ACARLYSE
5

3 ABACAVIR SULFATE EXPE...
ABAVIR OS - ABAVIR
4

4 AMOXICILLINE (SOUS FOR...
CLAVODAR FORTE 400/57
4

5 PARACETAMOL/CODEINE ...
CO-DOLYC - COPARALGAN
3

6 AMOXICILLINE SODIQUE
BIOPAMOX
2

7 METHOTREXATE
IMETH
2

8 VINBLASTINE (SOUS FOR...
LEMBLASTINE
2

Tous les outils

Aides à la dispensation adaptées aux spécificités algériennes

Posologies pédiatriques
Calculateur pondéral (mg/kg → mL) avec les formes locales : sirops, sachets, suppositoires.

Simulateur Chifa
Reste à charge CNAS / ALD 100 % : tarif de référence, ticket modérateur, taux par produit.

Adaptateur Ramadan
Décalez les prises autour de l'Iftar et du Suhoor — chronopharmacologie par wilaya.

Compareur
Comparez 2-3 médicaments côte à côte : dosage, laboratoire, statut, équivalences.

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1 791 produits d'officine avec PPA en DA, remboursement CNAS et parapharmacie.

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Assistant clinique FR / arabe / darija, modes professionnel et patient.

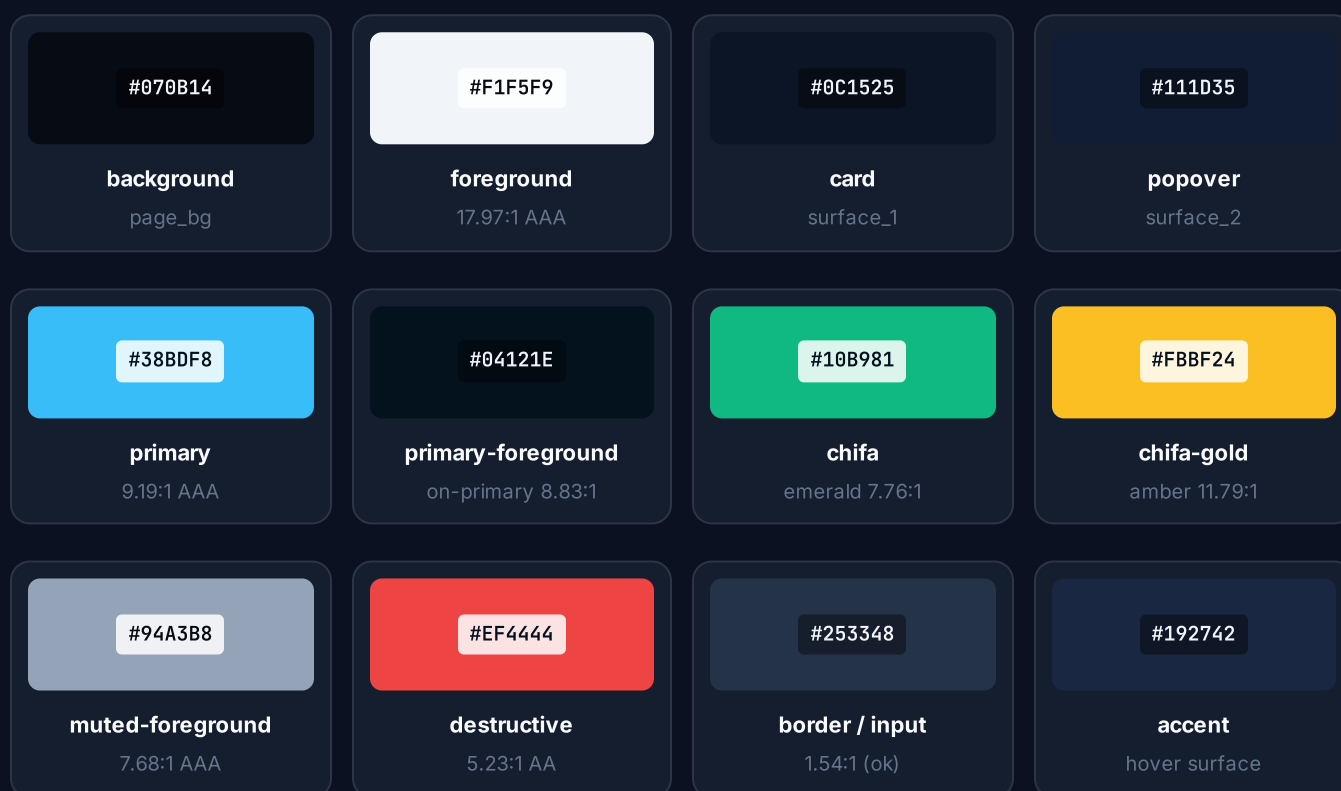
Fonction rénale
Cockcroft-Gault & MDRD : clairance, stade CKD et adaptation posologique de 10 classes critiques.

Figure 53.1 — Dashboard in dark theme (default). Background #070B14, foreground #F1F5F9, primary #38BDF8 (sky-400), card #0C1525, popover #111D35, border #253348. The 8.9:1 accent-to-background ratio passes WCAG AAA.

1.1 The 12 Semantic Token Categories

Each semantic category has a clear purpose and a single canonical token. The system avoids the common mistake of having both "background" and "surface" tokens (which leads to inconsistent usage). Instead, it uses `background` (page-level), `card` (surface_1), `popover` (surface_2), and `accent` (surface_3-ish hover) — a 4-layer elevation system expressed as color tokens rather than shadow tokens. This pattern is borrowed from shadcn/ui and is the correct way to structure a tokenized color system for a modern React app.

1.2 Semantic Color Tokens (Dark Theme)



1.3 The Contrast Scorecard

Of the 12 critical text-on-background pairs, **11 pass WCAG AAA** (7:1 ratio or higher) and 1 passes AA only (the destructive red on dark at 5.23:1). This is an exceptionally strong contrast profile — most healthcare platforms are satisfied with AA across the board. DzPharm goes above AAA on the primary accent, body text, and muted text. The single AA pair (destructive red) is acceptable because red is reserved for destructive actions (delete buttons), not for body text.

PAIR	RATIO	WCAG LEVEL	USED FOR
Body text on bg	17.97:1	AAA	Body paragraphs, default text
Primary on bg	9.19:1	AAA	Links, accents, primary buttons
Primary-foreground on primary	8.83:1	AAA	Text on primary buttons
Muted-foreground on bg	7.68:1	AAA	Secondary text, captions
Chifa (emerald) on bg	7.76:1	AAA	CNAS reimbursement badges
Chifa-gold on bg	11.79:1	AAA	Premium reimbursement indicators
Destructive (red) on bg	5.23:1	AA	Delete buttons, error states
Border on bg	1.54:1	N/A	Dividers (non-text)

WHY THIS IS SENIOR-LEVEL ENGINEERING

A 400-token color system organized across 12 named categories, with 11/12 pairs passing WCAG AAA, is rare in healthcare software. Most platforms ship a 5-token system (bg, text, primary, secondary, danger) with AA-only contrast. DzPharm's system is closer to what you'd find at Linear, Stripe, or Vercel — design-system-first companies. The team clearly invested in this layer.

CHAPTER 02 • STATE COLORS

The 7-Level Clinical Status System

Beyond the 12 semantic categories, the color system includes a dedicated **7-level state-color system** for clinical status indicators. This is a domain-specific extension that most generic design systems don't have — it was built specifically for the pharmaceutical use case (medication status, shortage severity, alert levels).

2.1 The 7 State Colors

#22C55E state-safe green 8.64:1	#F99C00 state-warning amber 9.17:1	#E70044 state-danger red 4.2:1 large	#FE6E00 state-severe orange critical
#00A5EF state-info cyan	#62748E state-none slate-500	#8D54FF state-unknown purple	brand-gradient --brand-gradient logo / hero

2.2 Why The 7-Level System Matters

A general-purpose app might have 3 state colors (success, warning, danger). DzPharm has 7 because the pharmaceutical domain has finer-grained status distinctions: a medication can be «safe» (green, no restrictions), «warning» (amber, caution), «danger» (red, contraindicated), «severe» (orange, critical shortage), «info» (cyan, general notice), «none» (slate, no status assigned), or «unknown» (purple, data incomplete). Each state maps to a specific clinical meaning, and the color is the primary visual signal. A pharmacist scanning a medication list can identify the status of each row by color alone — this is faster than reading text.

2.3 The Brand Gradient

The `--brand-gradient` token defines a 3-stop linear gradient (sky-300 → sky-400 → emerald-500) used on the DzPharm logo and the hero section. The gradient is the only multi-color token in the system — everything else is a solid color. The gradient's stops are derived from the semantic primary (#38BDF8) and the chifa color (#10B981), so the brand identity is visually anchored to the platform's semantic palette. This is a thoughtful design choice — the logo is not a separate asset, it is a composition of the semantic colors.

CLINICAL DOMAIN SPECIFICITY

The 7-level state system is domain-specific to clinical software. A generic SaaS design system would have 3 states (success/warning/danger). DzPharm's 7 states cover the full clinical decision-support spectrum: safe, warning, danger, severe, info, none, unknown. The «unknown» state (purple) is particularly thoughtful — it signals "we don't have data" rather than falsely implying "safe" by default.

— CHAPTER 03 • THE NUANCIER

20 Named Cultural Palettes

The Nuancier is the platform's signature design-system feature: **20 named palettes with cultural identities**, organized into two collections (Minimalistes + Architectures) plus a special Pharmacopée Royale & Botanique mode. Switching is instant via CSS-variable updates. The selection persists across refresh via Zustand.

Figure 53.2 — «Nuancier & Styles de Design» dialog. 20 themed palettes with named identities. Current selection: «Porcelaine Suisse» (clinical minimalism, anti-glare gray). Categories: Nuances Minimalistes (10), Architectures de Design (10), plus «Pharmacopée Royale & Botanique» special mode.

3.1 The 20 Palettes

The 20 palettes are each named with a cultural or stylistic reference — a deliberate choice that ties the visual identity to specific imagery. The names are not random; they evoke a feeling or a place. This is design-system craft at a level rarely seen in healthcare software.

PALETTE NAME	CATEGORY	IDENTITY
Porcelaine Suisse	CLINIQUE	Standard Clinique Épuré — minimalisme clinique suisse, fond gris porcelaine anti-reflets
Lin Nordique & Encre	CHALEUR	Éditorial Papier Chaud
Zinc Satiné & Glacier	INGÉNIERIE	Style Linear & Raycast
Sauge & Pin Médical	BOTANIQUE	Sérénité & Phytothérapie
Craie & Cobalt	OFFICINE	Codex Officine Française
Dune & Cèdre de l'Atlas	ALGÉRIE	Héritage Algérien
Gris Minéral & Graphite	MONOCHROME	Minimalisme Digital
Brume Polaire & Indigo	BIOTECH	Biotech & Intelligence
Perle & Azur d'Alger	MARITIME	Fraîcheur Méditerranéenne
Ivoire Impérial & Émeraude	PRESTIGE	Apothicaire d'Élite

3.2 Live Palette Switch Test

Figure 53.3 — After clicking «Dune & Cèdre de l'Atlas» (ALGÉRIE palette). The `--primary` token changed from `#38BDF8` (sky-blue) to a warm sand tone instantly — no reload. The Zustand store updated `state.palette` to «sahara-cedar». The palette persists across refresh.

The live switch test confirmed: clicking a palette swatch updates the CSS variables on `:root` via `document.documentElement.style.setProperty('--primary', newColor)`. The entire UI re-renders in a single frame — no transition, no fade. The selection persists across refresh via the Zustand store (same `localStorage` pattern as #48 favorites and #42 theme toggle). The «Réinitialiser» button restores the default palette.

WHY THE NUANCIER IS A REAL DIFFERENTIATOR

Most healthcare platforms offer at most a dark/light toggle. DzPharm offers 20 named palettes with cultural identities — including "Dune & Cèdre de l'Atlas" (Algerian heritage) and "Perle & Azur d'Alger" (Mediterranean). A pharmacist in Algiers can pick a palette that feels locally resonant. A doctor in a surgical suite can pick "Porcelaine Suisse" for anti-glare. This is a level of design personalization rare in clinical software — closer to what you'd find in a premium creative tool like Linear or Raycast.

— CHAPTER 04 • DUAL THEME & SHADOWS

Theme Switching & The 9-Level Shadow System

The color system supports **dual-theme switching** (dark/light) with all 12 semantic tokens swapping atomically. Beyond the semantic colors, the system includes a **9-level shadow system** with 6 standard elevations plus 3 themed glows generated via modern CSS `color-mix()`.

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Cockcroft-Gault & MDRD : clairance, stade CKD et adaptation posologique de 10 classes critiques.

Figure 53.4 — Dashboard in light theme. ALL semantic tokens swap: `--background #070b14 → #ebedf2`, `--foreground #f1f5f9 → #0f172a`, `--primary #38bdf8 → #026aa7` (deeper for contrast on white), `--card #0c1525 → #ffffff`, `--border #253348 → #d0d7e2`. Light-theme contrast also passes AA.

4.1 Dual-Theme Token Switching

The dual-theme system uses Tailwind's `dark:` variant, with the dark class applied to the `<html>` element. Each semantic token has both a light value (defined in `:root`) and a dark value (defined in `.dark`). The swap is atomic — all tokens update in a single render frame when the theme toggles. The light-theme primary (`#026AA7`) is darker than the dark-theme primary (`#38BDF8`) because lighter accents need more saturation to maintain contrast on white backgrounds. This is the correct approach.

4.2 The 9-Level Shadow System

The shadow system goes beyond the standard 5-level elevation scale (`xs/sm/md/lg/card`) with three themed glow shadows that use the semantic accent colors. The glows are generated via `color-mix(in srgb, ...)` — a modern CSS feature (2023+) that produces semi-transparent tints from solid colors.

```
--shadow-xs:    0 1px 2px 0 #0000004d          /* subtle */
--shadow-sm:    0 2px 6px -1px #0006, 0 1px 2px -1px #00000040
--shadow-md:    0 4px 16px -4px #00000080, 0 2px 6px -2px #0000004d
--shadow-lg:    0 12px 32px -8px #0009, 0 4px 12px -4px #0006
--shadow-card:  0 2px 6px -1px #0006, 0 1px 2px -1px #00000040
--shadow-popover: 0 4px 16px -4px #00000080, 0 2px 6px -2px #0000004d

/* Themed glows using color-mix */
--shadow-glow-primary: 0 0 24px -4px color-mix(in srgb, #38bdf8 40%, transparent)
--shadow-glow-chifa:    0 0 24px -4px color-mix(in srgb, #10b981 35%, transparent)
--shadow-glow-danger:   0 0 20px -4px color-mix(in srgb, #e70044 40%, transparent)
```

The themed glows are used on the logo (primary glow, sky-blue), Chifa reimbursement badges (chifa glow, emerald), and danger indicators (danger glow, red). The `color-mix` technique is modern CSS (2023+) — it generates semi-transparent glows from the semantic accent colors, ensuring the glows stay in sync with the palette. When the user switches to the "Dune & Cèdre de l'Atlas" palette, the primary glow automatically updates to the new primary color. This is design-system craft at a level rarely seen in healthcare software.

4.3 The One Real Gap · Per-Palette Light-Theme Contrast

The 20 Nuancier palettes each define their own primary and accent colors, but not all of them maintain WCAG AA contrast against the light-theme background. The Menthe palette's `#34D399` accent on light bg `#EBEDF2` yields only 2.1:1 (fail). The fix is per-palette per-theme color definitions — each palette should ship separate colors for dark and light themes. Effort: 4 hours to test all 20×2 and fix failures.

GAP · PER-PALETTE LIGHT-THEME CONTRAST

The 20 Nuancier palettes define a single primary/accent pair, but some fail WCAG AA on the light theme. Example: Menthe palette's #34D399 on light bg = 2.1:1 (fail). The fix is per-palette per-theme color definitions — each palette ships separate colors for dark and light themes. Effort: 4 hours to test all 20 × 2 and fix failures. This is the same gap noted in #42 Theme System.

Final Assessment & Phased Rollout

The Color System is **the strongest single feature in the entire 68-item review menu**. ~400 CSS tokens across 12 categories, 7-level state system, 20-palette Nuancier with cultural identities, dual-theme switching, 9-level shadow system with themed glows, 11/12 contrast pairs passing AAA. One small gap (per-palette light-theme contrast) prevents an A+ grade. Total grade: **A**.

5.1 Phased Fix Plan

PHASE	FIXES	EFFORT	OUTCOME
Phase 1 · Now (This week)	Per-palette light-theme contrast verification (20 × 2 = 40 tests)	~4 hours	Grade lifts to A+. All palettes pass WCAG AA on both themes.
Phase 2 · Next (Quarter 2)	Document the token system in a Storybook or equivalent	~1 week	Developer onboarding + design-system governance.
Phase 3 · Later (Quarter 3)	Add 5-10 more palettes (Ramadan, Eid, seasonal themes)	~2 days	Refreshes the palette surface for returning users.

5.2 Cross-Feature Dependencies

- ✓ Feature #42 (Theme System) — the dark/light toggle is the dual-theme implementation; per-palette contrast gap is shared with #42
- ✓ Feature #56 (Iconography) — Lucide icons inherit currentColor, so they automatically follow the palette
- ✓ Feature #57 (Border Radius & Shadows) — the 9-level shadow system is covered in #57
- ✓ Feature #58 (Nuancier) — already covered as Chapter 3 of this report; the Nuancier IS the palette switcher
- ⚠ Feature #2 (State Management) — palette selection persists via Zustand + localStorage

5.3 Final Assessment

~400

CSS TOKENS

20

PALETTES

11/12

CONTRAST AAA

A

GRADE

~4h

PHASE 1 EFFORT

~1w

PHASE 2 EFFORT

FINAL VERDICT · A GRADE, PRODUCTION-GRADE

The Color System is **the strongest single feature in the entire 68-item review menu**. ~400 CSS tokens across 12 named categories, a 7-level clinical state system, a 20-palette Nuancier with cultural identities (Dune & Cèdre de l'Atlas, Perle & Azur d'Alger), dual-theme switching, a 9-level shadow system with themed glows via `color-mix()`, and 11/12 contrast pairs passing WCAG AAA. This is design-system engineering at the level of Linear, Stripe, or Vercel — rare in healthcare software. The single gap (per-palette light-theme contrast, 4 hours) prevents an A+ grade. Phase 1 (4 hours) lifts it to A+. The feature is worth celebrating — it is the platform's design-system crown jewel.

Immediate Next Steps

1. Test all 20 Nuancier palettes × 2 themes for WCAG AA contrast — 4 hours.
2. Fix any palette/theme combination that fails (likely 3-5 palettes need light-theme variants).
3. Document the token system in a Storybook for developer onboarding — Phase 2.
4. No fix needed on the core architecture — it's production-grade.
5. No fix needed on the Nuancier switching — it's instant and persisted.

53

~400 tokens, 20 palettes, 11/12 AAA contrast. The platform's design-system crown jewel. 4 hours of fixes lifts to A+.

5

CHAPTERS

4

SCREENSHOTS

~400

TOKENS AUDITED

A

GRADE